

Aswin K

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Experience (3 years)

Webandcrafts

Jan 2024 – Present

Software Engineer

- Python Development Frameworks – Proficient in Python, with expertise in Django and FastAPI for building scalable, high-performance web applications.
- Contributed to the development of an all-in-one ERP application integrating sales, client management, accounts, leads, projects, and HR management functionalities.
- Implemented asynchronous task processing using RabbitMQ, handling 50K+ background jobs daily for email notifications, report generation, and data synchronization across microservices.
- Technical Research and Problem-Solving - Skilled in conducting in-depth research to evaluate technologies and implement innovative solutions. Adept at analyzing complex requirements, identifying best practices, and applying data-driven insights to enhance system efficiency and scalability.

Python Bootcamp

Jan 2023 – Dec 2023

Software Engineer Intern

- Developed and maintained RESTful APIs using Python and FastAPI, improving application response times and enhancing overall system performance.
- Built scalable web applications with Django framework, implementing MVC architecture and database optimization techniques for efficient data management.
- Designed and integrated NoSQL database solutions using MongoDB, optimizing data retrieval processes and supporting flexible schema requirements.
- Deployed applications on AWS cloud platform utilizing EC2 services, ensuring high availability and scalable infrastructure management.

Projects

Enterprise Business Management System | Python, Django, PostgreSQL, MongoDB, Redis

- Engineered scalable enterprise business management system using Python and Django framework, architecting full-stack solution for comprehensive business operations management.
- Integrated multiple business modules including sales management, client relationship management, accounting systems, lead tracking, project management, and HR functionalities into unified platform.
- Implemented high-performance data architecture utilizing PostgreSQL for relational data, MongoDB for document storage, Redis for caching, and Elasticsearch for advanced search capabilities.
- Designed resilient system architecture ensuring scalability and reliability for enterprise-level operations, supporting concurrent user access and real-time data processing across multiple business domains.

Ticketing System | Python, Django, RESTful APIs, Firebase, RBAC

- Developed backend architecture for admin dashboard and mobile application to manage tenant and worker requests efficiently.
- Designed and implemented RESTful APIs to facilitate seamless communication between users and the support system.
- Integrated role-based access control (RBAC) to ensure secure handling of tickets, assignments, and resolutions.
- Implemented real-time notifications via Firebase Cloud Messaging (FCM) to improve response times and service efficiency.

LLM-POWERED VOICE AGENT - NEST MATRIMONY | FastAPI, Gemini, LiveKit, MongoDB, Google STT/TTS

- Designed and developed RESTful APIs using FastAPI to manage users, matchmaking data, and call workflows.
- Implemented a Gemini-powered conversational AI agent to explain matching profiles and user interests in natural language.
- Built a real-time voice interaction system using LiveKit for low-latency audio streaming.
- Integrated Google Speech-to-Text (STT) for real-time user speech recognition during calls.
- Implemented outbound AI voice calling logic to proactively notify users about compatible matches.
- Integrated Google Text-to-Speech (TTS) to generate natural, human-like voice responses.

Technical Skills

Languages: Python, JavaScript, SQL
Backend Development: Django, FastAPI, RESTful API Design, Microservices Architecture, Asyncio
Frontend Development: React.js, JavaScript ES6+, HTML5, CSS3, Responsive Design
Databases: PostgreSQL, MongoDB, Redis, MySQL, Database Design, Query Optimization, Indexing
Cloud & Infrastructure: Amazon Web Services (AWS EC2, S3, RDS, Lambda), Docker, Linux
DevOps & CI/CD: Git, GitHub, Docker, CI/CD Pipelines, GitHub Actions, Automated Deployment
Message Queues & Caching: RabbitMQ, Redis, Celery, Asynchronous Task Processing
Payment Integration: Stripe API, Webhook Handling, Payment Processing, Subscription Management
AI & Automation: LangChain, OpenAI API, Crew AI, AI Agents, Vector Databases, Prompt Engineering, n8n, Make
API & Integration: RESTful APIs, Third-Party API Integration, Postman, API Documentation
Security: Role-Based Access Control (RBAC), JWT Authentication, OAuth, Data Security
Architecture & Design: Microservices, System Design, Database Architecture, Scalable Systems, MVC Pattern
Performance Optimization: Caching Strategies, Query Tuning, Async Programming, Load Optimization
Tools & Methodologies: Jira, Agile, Scrum, Code Review, Version Control, Testing

Education

University of Calicut	2019-2022
<i>Bachelor of Computer Application (BCA)</i>	